The Modern AI System Map

T1-01

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A model is not the system

A product combines intent, data, a trained model, runtime orchestration, people, and governance. A model score alone does not establish product value or safety.

Five planes

Intent and policy $\rightarrow$ data and training $\rightarrow$ model and adaptation $\rightarrow$ serving and interaction $\rightarrow$ measurement and governance. Every arrow is an evidence handoff.

Data handoff

Record provenance, permissions, transformations, splits, and gaps. Ask: does the test set represent the decision we want to make?

Model handoff

Version the weights, code, prompt, and adaptation. Pre-training and post-training alter distributions of behavior; neither creates a guarantee of truth.

Runtime handoff

The real path includes retrieval, tools, auth, caching, policy, UI, and logging. Evaluate this path—not only a bare model call.

Evaluation handoff

Use representative tasks, stress cases, error analysis, and calibrated uncertainty.
Aggregate accuracy hides important failure modes.

Course-help example

Restrict retrieval to released materials; cite sources; block hidden-solution leakage and write tools; test ambiguous questions and injection attempts.

Discussion

Where would you locate a wrong answer: corpus, prompt, tool, model, UI, or policy? What observation would discriminate among them?